

Building a universal game-playing agent infrastructure to optimize performance

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IM updates

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Research question: reminder

Goal: create a general dialogue game-playing agent that

- does not rely on game-specific solvers
- does not modify game-specific rules
- must adhere to the standard clembench game flow

Method: adding different modular components around vanilla LLMs (Reflect, ReAct, external databases access etc.)

Time plan (from last meeting)

November:

- 1) clembench games exploration (especially multimodal)
- 2) finalizing possible improvement paths (and their combinations)

December:

- 1) implementation and testing
- 2) refactoring and testing other paths if no improvements

Progress so far

- Clembench games have been explored (test runs with default instances using qwen3-coder-30b-a3b-Instruct-fp8), although multimodal ones were not tested yet
- They were clustered to track progress and results in the future (namely, to identify general vs. domain-specific problems across success/failure cases)

Key insights:

- 1) Only several games allow self-correction for the players. Thus, some standard agent-like architectures (e.g., reflection) are **not suitable** for certain games
- 2) **Think before you act** approach should work better in this setting, because invalid format can easily abort a game and thus leave no opportunity for reflection

Agent enhancements

Why start with Reflect?

- Helps to address errors (ParseError, GameError) reliably

Important: Give short, direct answers!

Let us start.

Player 1 (Answerer) to GM (Game Master)

Ok.

GM (Game Master) to Player 1 (Answerer)

ME: Does the questioner know what the number for a is? Answer this question with yes or no and then stop.

Player 1 (Answerer) to GM (Game Master)

No.

Game Master: parse

NA

GM (Game Master) to Player 1 (Answerer)

ME: Please answer this question carefully. Does the questioner know what the number for a is? Answer this question with yes or no and then stop.

Player 1 (Answerer) to GM (Game Master)

No.

Game Master: parse

NA

GM (Game Master) to Player 1 (Answerer)

ME: Please follow my instructions. Does the questioner know what the number for a is? Answer this

Agent enhancements: Reflection

Why start with Reflect?

- Helps to address errors (ParseError, GameError) reliably
- Provides a foundation for ReAct

Agent enhancements: Reflection

- When `add_player()` is called in `player.py`, an **ErrorReflectComponent** is created and attached to the player
- The reflection is made with the player's own model or a separate one
- While the game proceeds, if an error is caught, the reflection with a simple prompt is triggered:

"You just made a mistake. Analyze what went wrong:

In 2-3 sentences, explain:

1) What was the mistake?

2) How should you correct it?

IMPORTANT: Only provide your analysis in the "explanation:" format. Do NOT include a "guess:" in your reflection."

- After the reflection is generated, it's added to both `self._messages` and `perspective`, and thus, becomes a part of the input for the next response generation

Agent enhancements: Reflection testing

- wordle game
- default instances (30), no clue, no critic
- with and without reflection
- tested using qwen3-coder-30b-a3b-Instruct-fp8
- different sampling temperature

Agent enhancements: Reflection testing

- wordle game
- default instances (30), no clue, no critic

		t = 0.0	t = 0.8
with reflection	clemscore	0.66 	8.33 
	quality score	1.33 	20.83 
without reflection	clemscore	6.66	2.67
	quality score	7.69	6.67

Agent enhancements: Reflection testing

- wordle game
- default instances (30), no clue, no critic

		t = 0.0	t = 0.8
with reflection	Average Closeness Score	7,47 	9,67 
	Average Strategy Score	1,75 	16,37 
without reflection	Average Closeness Score	8,85	8,9
	Average Strategy Score	12,75	8,4

Reflecting on Reflection

- Reflection + lower temp $\Rightarrow$ lower performance
- Reflection + higher temp $\Rightarrow$ higher performance
- Using a different model for reflection might improve the performance, even with low temperature
- It remains unclear how to handle the games where self-correction is not allowed